

Ömer Sayılğan

Roll No.: 03796208

Master of Science

Technical University of Munich

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🐙 GitHub Profile

🌐 LinkedIn Profile

EDUCATION

- **Master of Science in Aerospace Engineering** 2024-2026
Technical University of Munich, Munich/Germany CGPA: 2.6
- **Bachelor of Science in Mechanical Engineering** 2017-23
Middle East Technical University, Ankara/Turkey CGPA: 3.33/4.0

PERSONAL PROJECTS

- **Ball Balancing Robot**
Automatic ball balancing system using 3D printed parts, servo motors and Pi Pico W
 - Implemented PID control for servo motors.
 - Designed parts using 3D CAD and produced them via 3D printing.
 - Achieved wireless communication via MQTT using Pi Pico W.
- **My Own Website: Link to My Website**
Created a personal website using HTML and CSS
 - Technologies Used: HTML, CSS, JavaScript, GitHub.
- **Encoder Design Using Photoresistor**
Implemented a basic encoder using a photoresistor sensor
 - Used Pi Pico W for wireless communication of measured angles.
 - Photoresistor used as the angle sensor.

EXPERIENCE

- **Control Systems Engineer** Aug 2023 - May 2024
ROKETSAN
 - Created Simulink models of missile seeker.
 - Designed a robust controller using data driven identification.
 - Developed algorithms on Simulink and tested it via Simulink Real Time using xPC.
 - Created a GUI to automatize the calibrating process for technicians.
 - Hands-on experience with calibrating the sensors such as gyros and resolvers.
- **Mechatronics Engineer Internship** Aug 2022 - Sep 2022
ROKETSAN
 - Learned the fundamentals of Inertially Stabilized Platforms (ISP).
 - Learned basics of seeker control.
 - Recreated an algorithm to transform the encoder rotations to angular measurement.
 - Created a basic 3D CAD model of missile seeker and selected ball bearings.
 - Calculated stresses on launch in SOLIDWORKS.

TECHNICAL SKILLS AND INTERESTS

Languages: C/C++, Python, MATLAB, LATEX, HTML+CSS

Libraries : C++ STL, Python Libraries, MATLAB Robust Control Toolbox

Engineering Programs: Simulink, Siemens NX 11, SMath, MATHCAD, Xilinx IDE

Relevant Coursework: Flight Dynamics, Signals and Systems, Control Engineering, Digital Signal Processing, Introduction to ROS, Model Reference Adaptive Control, Mechatronic Instrumentation.

Areas of Interest: Control systems design and identification, Embedded systems.

Soft Skills: Problem Solving, Self-learning, Presentation, Adaptability

HOBBIES AND INTERESTS

- Multi-instrumentalist, played guitar and did vocals in a rock band.
- Chess and go enthusiast.
- Loves to travel.